

Heat Insurance at Work

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November 4, 2025

Climate change is causing extreme heat now

- ▶ In India in 2024:
 - 37 cities surpassed 45° C (113° F)
 - 500 out 741 districts hit 40° C at least once in April/May
 - 3 districts in Maharastra hit 40° C on 48 of 61 days.
- ▶ Extreme heat has negative physical, economic and emotional impacts (e.g. Burgess et al. 2017)
- ▶ Should social protection systems change to reflect new climate risks?

This paper

- ▶ RCT of parametric heat insurance: pays daily wage when temperature passes threshold (41.6°C)
 - Highly subsidised (pay INR300, get INR400 + heat payments)
 - Implemented by the Self-Employed Women's Association (SEWA) in India: 7 states 250,000 women
 - Randomise recruitment incentives at the village level to generate experimental variation in take-up
- ▶ Study impact on behaviours (adaptation, coping strategies) and welfare (health, mental health).
 - Baseline data & participation from SEWA membership database
 - Outcomes measured through surveys in Sept-Nov 2025
- ▶ Big picture: Can it exist without subsidies? If not, is it a good use of public / philanthropic funds?

Should the government do anything about extreme heat?

- ▶ Extreme heat is jointly an (i) income shock (ii) expenditure shock.
 - Can't work, lose earnings
 - Get sick, need healthcare
- ▶ How do we deal with risk as individuals
 - Diversify risk (e.g. heat insurance, health insurance, informal insurance)
 - Take actions to reduce exposure/damages (e.g. use aircon / drink water / reduce work hours)
- ▶ Why might government intervene:
 - Missing insurance market → support (make obligatory, subsidise) or provide (social insurance)
 - Inefficiently low adaptation → information / subsidies / fix credit markets

Why might subsidised heat insurance be a good idea?

Suppose this product isn't financially viable without subsidies

- ▶ Hard to eliminate risk with exposure reduction alone.
- ▶ Why not use (i) unemployment insurance (ii) health insurance (iii) redistributive programmes
 - UI and health insurance don't typically cover informal workers
 - Redistributive programmes don't respond dynamically
- ▶ Maybe this is more feasible? Why do we ever target particular risks?
- ▶ Key climate adaptation question: do something different or make existing systems work better.

The Self-Employed Women's Association has 3.2 million members



Vendors and Hawkers



Home-based Workers



Labour & Service Providers



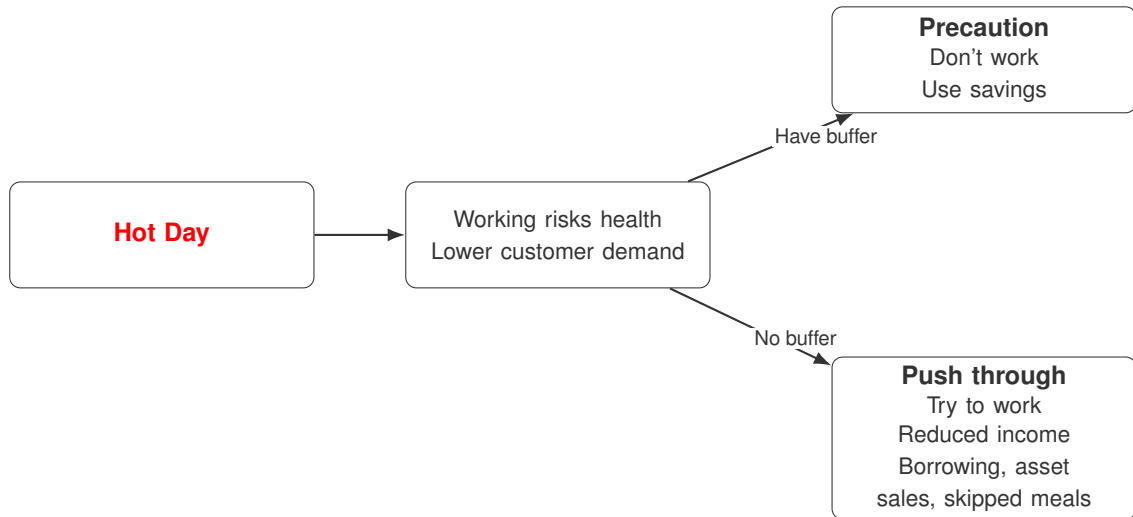
Producers (e.g. farmers)

Qualitative impacts of extreme heat on SEWA members

Consistent with empirical literature

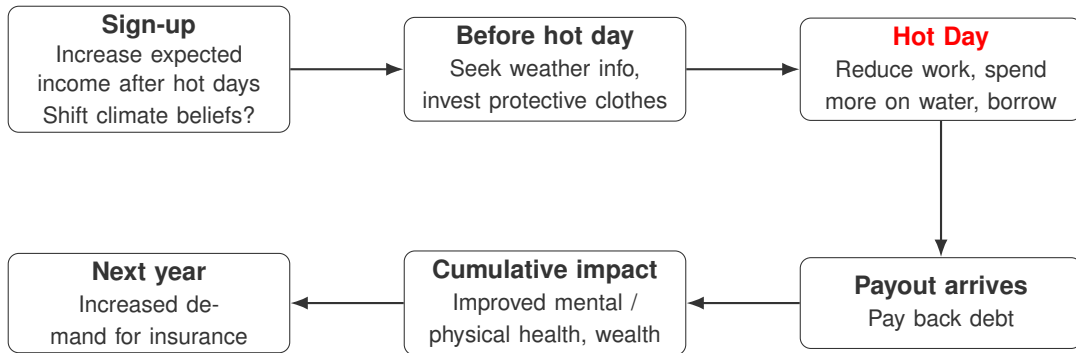
- ▶ Heat-related health impacts
 - headaches, dizziness, fainting, vomiting, fungal infections, muscle cramps, rapid pulse, fever, UTIs
 - Exacerbated by lack of access to drinking water, sanitation, and cooling (e.g. aircon)
- ▶ Reduced labour productivity + lost income:
 - Severe health consequences from working during hottest hours of the day
 - Spoilage of produce, loss of customers, transport disruption
- ▶ Loss of income → financial and emotional distress
 - High interest informal loans, sale of productive assets
 - Mental stress of financial challenges and increased care responsibilities

Responses to a hot day



- ▶ **A promise of cash transfers after hot days**
- ▶ **Actual liquidity in womens' bank account 4-6 weeks after a hot day**
- ▶ A reminder about the health and livelihood risks associated with extreme heat
- ▶ A signal that a threshold exists beyond which conditions are especially hazardous
- ▶ A message of support - conveying that “someone is looking out for me” during climate stress

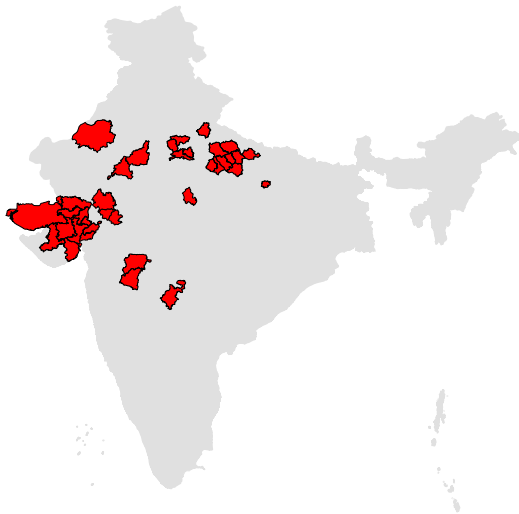
A Theory of Change



RCT design

- ▶ All SEWA members in study areas are eligible for the heat insurance programme.
- ▶ Grassroot leaders recruit members in their assigned villages.
- ▶ To generate exogenous variation in take-up:
 - Villages are randomly assigned to treatment and added to a “priority list.”
 - Leaders receive incentives to recruit only in priority (treatment) villages.
- ▶ Use village-level treatment assignment as an IV for individual participation.

Study covers 5,631 villages in Gujarat, UP, Rajasthan, and Maharashtra



Key outcomes

- ▶ **Use of exposure-reduction strategies on hot days** e.g. adjusting work hours, hydration.
- ▶ **Use of financial coping strategies** e.g. skipping meals, borrowing, or selling assets.

Secondary outcomes

- ▶ Heat-related health problems
- ▶ Healthcare use (expenditure and visits)
- ▶ Assets and liabilities
- ▶ Consumption and income
- ▶ Women's empowerment
- ▶ Climate beliefs and willingness to pay for insurance
- ▶ Understanding of insurance mechanisms
- ▶ Spillovers on household members (e.g., labour supply, education)

Heterogeneity analysis (examples)

► **Occupation:**

- Differences in workplace, conditions and flexibility impact (i) heat exposure (ii) avoidance potential.
- Test for heterogeneity by occupational group, 'heat exposure' and market power.

► **Access to credit and savings:**

- Insurance may benefit those lacking financial buffers the most.
- Test for heterogeneity by baseline income and local credit market access.

► **Prior exposure to heat events:**

- Past heatwave exposure may shift perceptions and responses.
- Construct weather histories from secondary data.

Number of payments RDD

- ▶ Leverage payout threshold to implement regression discontinuity design (RDD).
- ▶ Compare “just eligible” vs. “just ineligible” by simulating eligibility under slightly adjusted thresholds.
 - **Control group:** villages that would have received an additional payout under a higher threshold.
 - **Treatment group:** villages that would have received one fewer payouts under lower threshold.